A Technical Guide to Logistic Regression

Optimisation, Model Fit & Regularisation

From gradient descent mechanics to taming overfitting — the core math every practitioner needs.

Sections: Gradient Descent Fundamentals • Optimisation Accelerators • Model Fit • Combating Overfitting • Regularisation Mechanics • Regularised Gradient Descent • Conclusion

1. Fundamentals of Logistic Regression Optimisation

The objective of fitting a logistic regression model is to identify the optimal parameters w (weights) and b (bias) that minimise the cost function J(w, b). In a clinical diagnostic environment, this allows us to input features x (such as tumour size or patient age) and output a prediction f(x) — the estimated probability that the label y is 1 (e.g., a diagnosis of malignancy).

The Gradient Descent Algorithm

To minimise the cost, we utilise the Gradient Descent algorithm. This process requires us to update our parameters iteratively. To implement this correctly, we must perform **Simultaneous Updates**, meaning we calculate the update values for all parameters before overwriting the old values.

EQUATION

$$w_j = w_j - \alpha \cdot \partial/\partial w_j \cdot J(w, b)$$
$$b = b - \alpha \cdot \partial/\partial b \cdot J(w, b)$$

Where α is the learning rate, n is the number of features, and m is the number of training examples.

Mathematical Derivatives

The derivative terms, which represent the gradient of the cost function, are calculated as follows:

EQUATION

$$\partial/\partial w_j \cdot J(w, b) = (1/m) \cdot \Sigma_{i=1}^{m} (f(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$
$$\partial/\partial b \cdot J(w, b) = (1/m) \cdot \Sigma_{i=1}^{m} (f(x^{(i)}) - y^{(i)})$$

Key Technical Distinction

It is a common point of confusion for students that these update equations look identical to those in Linear Regression. However, they represent two very different algorithms because the definition of f(x) has changed. In Linear Regression, f(x) = wx + b. In Logistic Regression, f(x) is the Sigmoid function applied to the linear term:

EQUATION

$$f(x) = 1 / (1 + e^{-(wx + b)})$$

2. Practical Enhancements for Gradient Descent

To ensure the optimisation process is both stable and computationally efficient, practitioners employ specific “accelerators.”

Optimisation Accelerators

Technique	Benefit to Gradient Descent
Feature Scaling	Scaling features to similar ranges (e.g., –1 to +1) allows gradient descent to find the path to the minimum more directly, significantly speeding up convergence.
Vectorisation	By replacing explicit loops with matrix operations, we leverage modern hardware architecture to execute calculations in parallel, reducing execution time.

3. The Spectrum of Model Fit: Underfitting vs Overfitting

The ultimate goal in machine learning is **generalisation**: the ability of a model to make accurate predictions on brand-new data it has never seen before. We often describe the search for this balance using the "Goldilocks" analogy.

States of Model Fit

1. **Underfitting (High Bias):** Think of this as the algorithm having a strong preconception about the data. For instance, if the algorithm is biased toward a linear relationship when the housing data clearly curves, it will fail to capture the pattern. It is "too cold" to be useful.
2. **Overfitting (High Variance):** Here, the algorithm is trying very hard to fit every single training example. This results in a complex, "wiggly" curve that passes through every point but is overly sensitive to noise. While it performs perfectly on training data, it fails to generalise. It is "too hot."
3. **"Just Right":** This is the ideal state where the model captures the underlying pattern (e.g., an elliptical decision boundary in tumour classification) without being distracted by individual outliers.

4. Strategies to Combat Overfitting

When a model suffers from high variance, three primary strategies can be employed:

1. **Collecting More Data:** This is the most effective tool. A larger dataset forces the algorithm to learn a smoother, more representative function.
2. **Feature Selection:** By choosing only a subset of the most relevant features (e.g., house size and age), we simplify the model. However, a major disadvantage is that the algorithm is throwing away information that might actually be useful.
3. **Regularisation:** This is often the superior choice because it allows us to keep all features while gently reducing their impact. It prevents any single feature from having an "overly large effect" on the outcome.

5. The Mechanics of Regularisation

Regularisation functions by adding a penalty for large parameter values to the cost function, encouraging the model to be "simpler" and "smoother."

The Regularised Cost Function

The modified cost function J(w, b) is expressed as:

EQUATION

$$J(w, b) = [\text{Original Cost}] + (\lambda / 2m) \cdot \Sigma_{j=1}^{n} w_j^2$$

A key technical detail is the division by 2m. Scaling both the error term and the regularisation term by the number of examples makes it significantly easier to choose a λ that remains effective even as your training set size (m) grows.

The Regularisation Parameter (λ)

The parameter λ controls the trade-off between fitting the training data and keeping the parameters small.

Lambda Value	Effect on Model
λ = 0	No regularisation; high risk of overfitting.
$\lambda \to \infty$	Parameters w are forced to ≈ 0. The model becomes f(x) ≈ b, appearing as a flat horizontal line (Underfitting / High Bias).
λ "Just Right"	Successfully balances training error with parameter shrinkage for optimal generalisation.

Note on Parameter b: By convention, we do not regularise the bias b. While you could include it, doing so makes very little practical difference in the model's performance.

6. Implementing Regularised Gradient Descent

In a regularised context, the update rule for w_j is modified to include the penalty term.

Modified Update Rule

EQUATION

$$w_j = w_j - \alpha \cdot [\, (1/m) \cdot \Sigma_{i=1}^{m} (f(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)} + (\lambda/m) \cdot w_j \,]$$

The "Shrinkage" Perspective

We can rewrite the math to see exactly what happens to w_j during each iteration:

EQUATION

$$w_j = w_j \cdot (1 - \alpha\lambda/m) - \alpha \cdot (\text{Standard Gradient Term})$$

In practice, the term (1 – αλ/m) is a value slightly less than 1 (for example, 0.9998). This means that on every single iteration, we "shrink" the value of w_j slightly before applying the standard update.

7. Conclusion for the Machine Learning Practitioner

While any coder can fit a model, the specific high-value skill that separates senior practitioners from juniors is knowing when and how to reduce overfitting. This intuition is what enables engineers in hubs like Silicon Valley to create massive value for their organisations.

In the industry, you will likely use professional libraries like Scikit-Learn to handle these implementations. Furthermore, the concepts of cost functions, simultaneous updates, and regularisation are the exact building blocks required for Neural Networks and Deep Learning. Mastering these foundations today is your first step toward building the advanced AI systems of tomorrow.

Takeaway: Regularisation lets you keep every feature while capping its influence — tune λ so the model fits the signal, not the noise, and you've mastered the exact building block that scales up to neural networks.